



Problem and Motivation

School of Computational Sciences and Artificial Intelligence, Faculty of Science and Technology - June 2026

Problem: given a chest X-ray, decide pneumonia vs normal, localize the region, and produce a report. Done manually this is slow and needs scarce radiologist expertise, and no accessible, explainable tool does it end to end.

Motivation: high X-ray volume and few radiologists delay reads; a missed pneumonia is the costly error; existing tools are closed and expensive.

We give a fast, explainable AI second reader, doctor in control.

Objectives

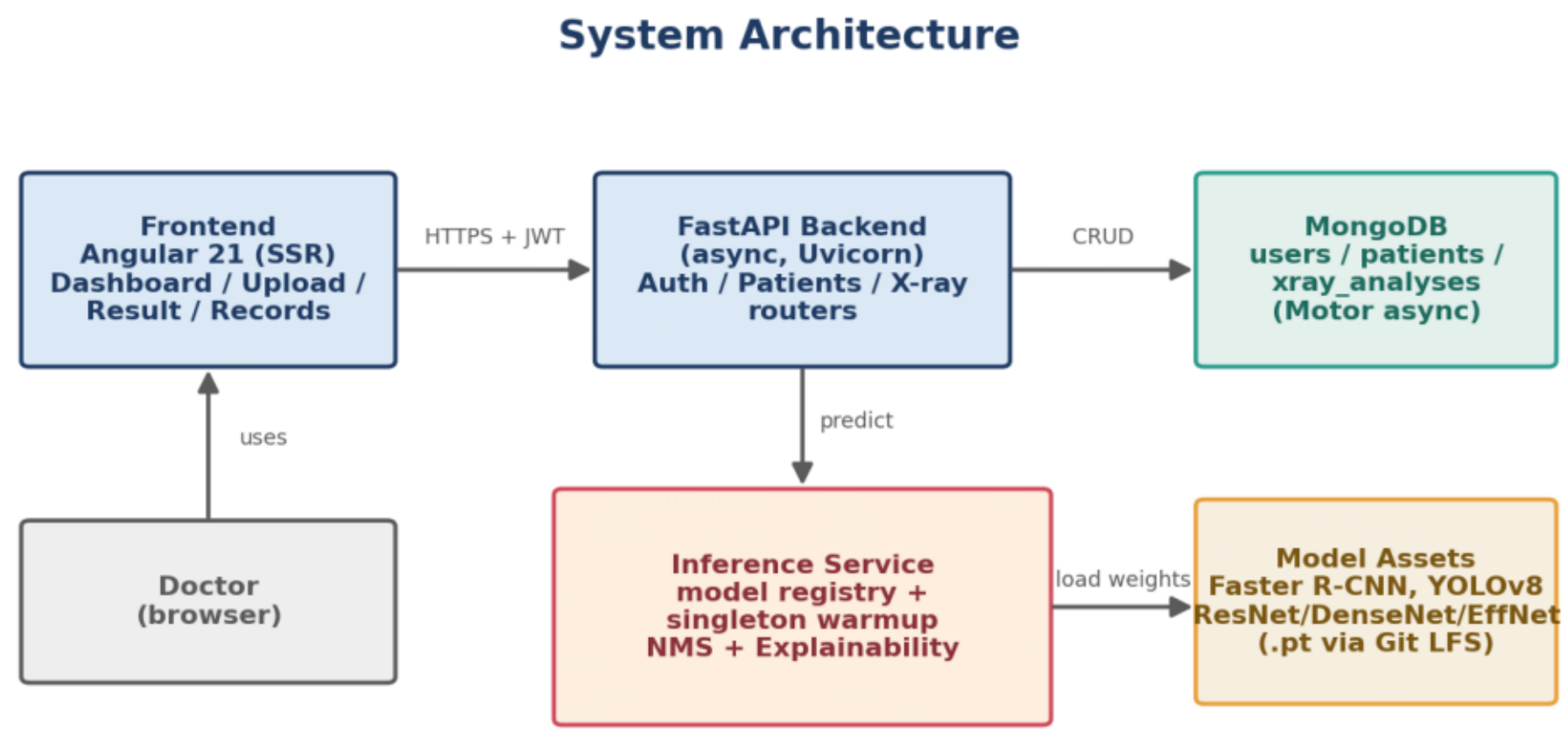
- Train and honestly evaluate models for pneumonia detection and localization.
- Deliver a secure, multi-user web app with patient management and history.
- Provide visual explainability with every prediction.
- Keep the system reproducible, documented, and maintainable.

Proposed Solution



A doctor signs in, uploads a chest X-ray, and gets an automated reading: a pneumonia decision with confidence, a localized bounding box, an explainability heatmap, and an auto-filled report saved per patient. Five models behind one registry (Faster R-CNN deployed); the physician confirms every result.

System Architecture



Technologies Used

Frontend: Angular 21 (SSR). Backend: FastAPI (async), Uvicorn.
Database: MongoDB (Motor). AI/ML: PyTorch, torchvision, Ultralytics.
Security: JWT, bcrypt, CORS. Model storage: Git LFS.

Key Numbers

0.886

Best AUC (EfficientNet-B0)

5

Deep-learning models served

0.812

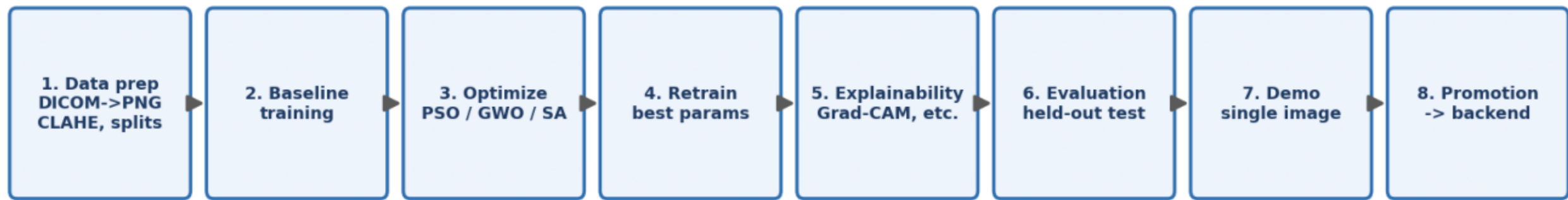
Recall (Faster R-CNN)

26,684

RSNA radiographs

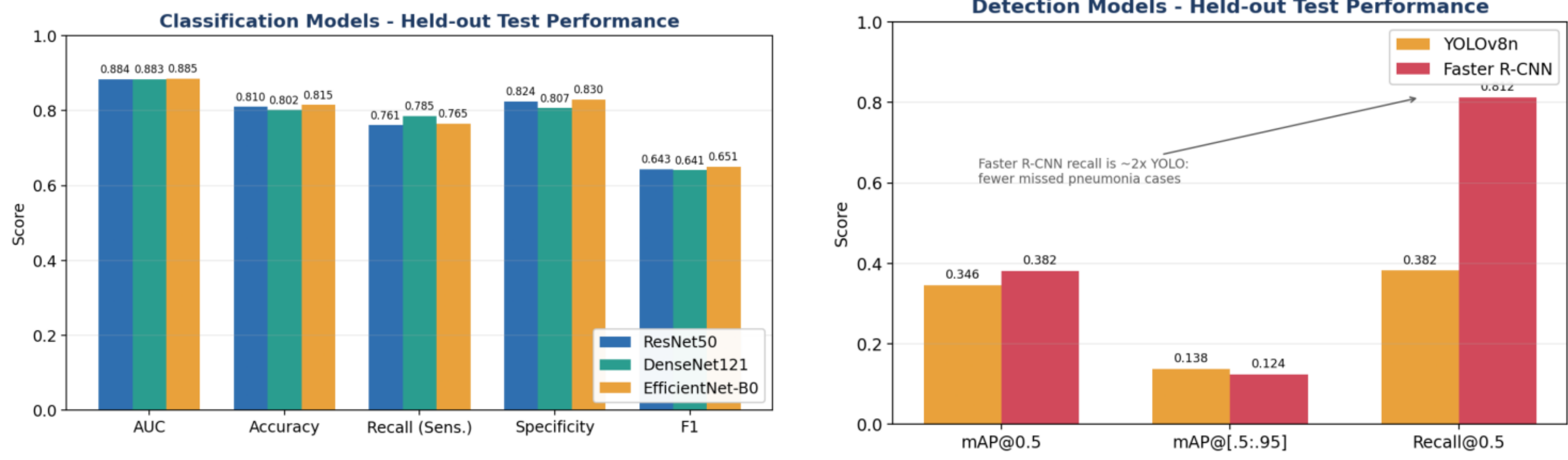
Methodology

AI / ML Pipeline (8 phases)



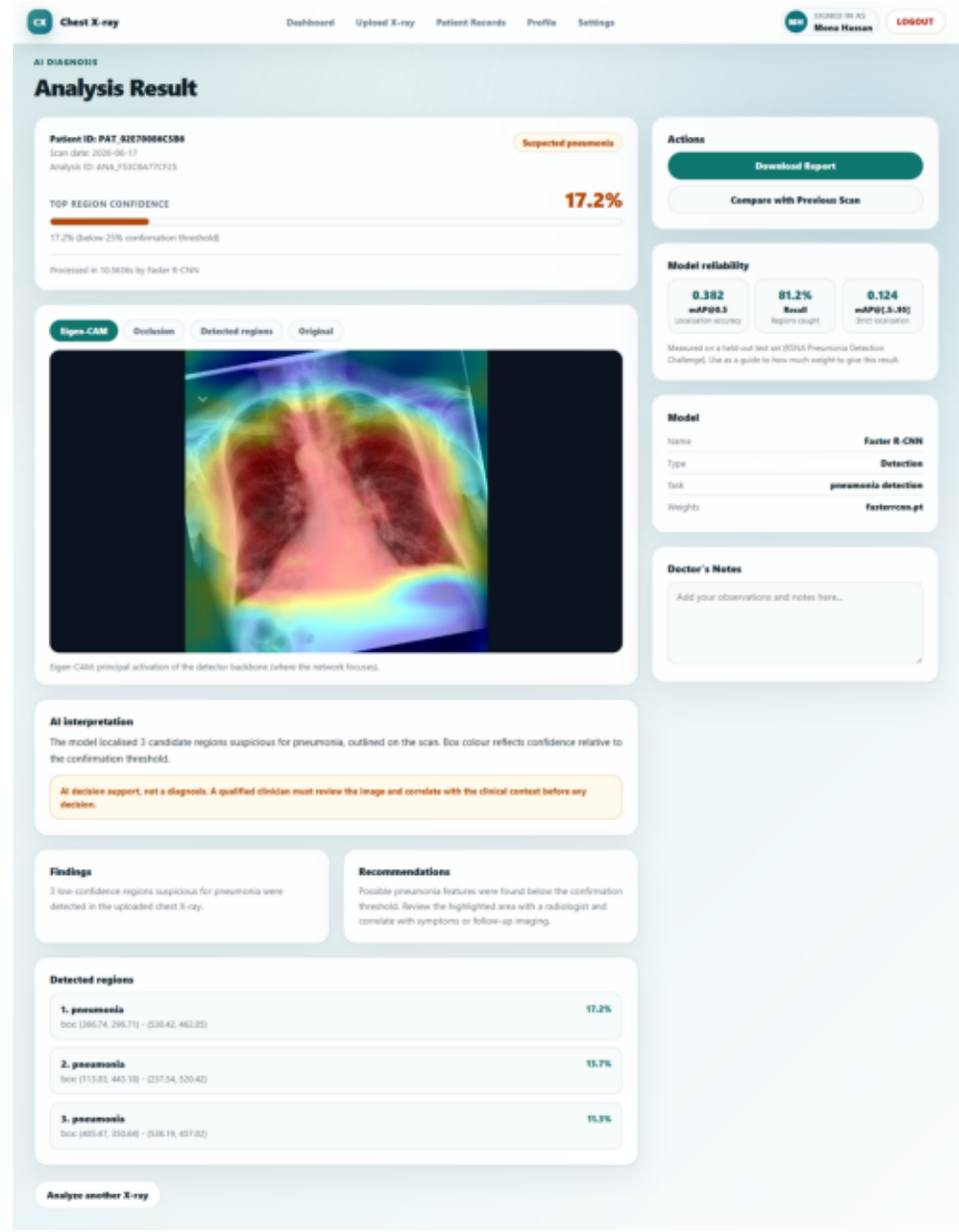
RSNA Pneumonia Detection Challenge (~26,684 radiographs), DICOM to PNG, leak-free patient-wise split (20% held-out test: 4,135 normal / 1,202 pneumonia). Transfer learning, mixed-precision training, augmentation, and early stopping; PSO/GWO/SA hyperparameter search.

Results (held-out test set)



Best classifier EfficientNet-B0: AUC 0.886. Deployed detector Faster R-CNN: recall 0.812 (mAP@0.5 0.381). Recall is prioritized because a missed pneumonia is the costly error. Numbers are honest and competitive with the literature (RSNA detection mAP typically 0.32-0.39).

Sample Output and the App



Left: detector output with the localized region. Right: the result screen - decision, confidence, bounding box, heatmap, and the auto-filled report. Every prediction shows a heatmap (Grad-CAM and related) so the clinician can verify the model looks at the lungs, not an artifact.

Conclusion

A complete, working, explainable full-stack pneumonia screening system: five models rigorously evaluated, secure and reproducible, demonstrating the full path from research model to usable clinical product.

